

ABHISHEK KUMAR SINGH

DevSecOps Engineer & SRE | Cloud & Web3 Security

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Cloud & DevSecOps Engineer specializing in autonomous, self-healing infrastructure and Web3 security. Proven expertise in architecting zero-trust cloud environments utilizing GitOps deployment strategies, shift-left security quality gates, and event-driven observability. Highly adept at reducing operational toil and Error Budgets through automated remediation pipelines across AWS and Kubernetes ecosystems. Operating with IST/SGT alignment for global and APAC teams.

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (EKS, EC2, VPC, IAM, S3, EventBridge), Terraform (IaC), Linux Architecture.
- **Containers & GitOps:** Kubernetes, Docker, Helm, Kustomize, ArgoCD Declarative Deployments.
- **DevSecOps & Security Engineering:** Snyk, Trivy, SonarQube, OIDC Identity Federation, IRSA, Zero-Trust Networking, Static Analysis (Slither), Formal Verification (Certora).
- **Observability & AIOps:** Prometheus, Grafana, CloudWatch Anomaly Detection, AlertManager; SLI Tracking (Four Golden Signals: Latency, Traffic, Errors, Saturation).
- **Automation & CI/CD:** GitHub Actions, Jenkins, Git, Python (Boto3), Shell Scripting (Bash).
- **Data Science & Analytics:** Statistical Analysis, Predictive Modeling, Pandas, Scikit-Learn (IIIT Bangalore Certified).

PROFESSIONAL EXPERIENCE

Cloud & DevSecOps Engineer (Professional Portfolio)

Remote | Jul 2025 – Present

- Architected zero-trust, multi-AZ cloud infrastructure on AWS utilizing modular Terraform, enforcing least-privilege network isolation and IRSA identity management.
- Engineered GitOps-driven CI/CD pipelines using GitHub Actions and ArgoCD with embedded shift-left security (Trivy, Snyk), achieving a 100% block rate on critical CVEs.
- Reduced infrastructure Mean Time to Recovery (MTTR) by 97% by developing event-driven AIOps remediation engines (AWS Lambda, Python Boto3) for self-healing Kubernetes clusters.
- Established proactive observability by instrumenting EKS environments with the Prometheus/Grafana stack, leveraging CloudWatch Anomaly Detection to minimize false-positive alerts by 40%.

Graduate Administrative Assistant

Troy University, Alabama, USA | May 2024 – Jun 2025

- Optimized and administered the WC Online Portal database to streamline scheduling and payroll workflows, ensuring 100% data integrity for university-wide tutor operations.
- Co-facilitated technical workshops on Generative AI alongside university faculty, translating complex software concepts into engaging, accessible training for diverse audiences.
- Provided advanced mentorship to Computer Science undergraduates via code reviews, while demonstrating high emotional intelligence and cross-functional collaboration during university-wide orientation events.

President, Indian Student Association

Troy University, Alabama, USA | May 2024 – Jun 2025

- Led a cross-functional executive board to manage organizational budgets, logistics, and operational execution for large-scale university events and cultural programs.
 - Acted as the primary liaison between the university's international student body and local government, advocating for community integration initiatives directly to the Mayor at the annual Troy City Town Hall.
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ENGINEERING PROJECTS

DevSecOps CI/CD & GitOps Infrastructure

GitHub Actions · ArgoCD · Kustomize · AWS EKS · Prometheus · Grafana | [GitHub Link](#)

- Architected an end-to-end declarative GitOps pipeline to securely build, deploy, and monitor containerized applications on AWS EKS.
- Enforced strict shift-left security by integrating Trivy and Snyk for automated CVE blocking, alongside passwordless AWS OIDC authentication.
- Instrumented the Kubernetes environment with the Prometheus and Grafana stack, configuring ServiceMonitors for dynamic target discovery and real-time observability.

Smart Score CLI & Web3 DevSecOps Action

Python · Docker · GitHub Actions · Pydantic · Hardhat | [GitHub Link](#)

- Developed a specialized CLI tool that scores Solidity smart contracts on security and efficiency metrics, successfully translating rigorous academic thesis research into deployable software.
- Engineered a multi-stage Docker build to minimize container footprint and integrated the tool as a reusable GitHub Action to act as an automated CI/CD quality gate.

Event-Driven AIOps Remediation Engine

AWS Lambda · EventBridge · CloudWatch Anomaly Detection · Python Boto3 | [GitHub Link](#)

- Architected an event-driven framework that acts as a serverless Kubernetes operator to automate incident response and node recovery workflows.
- Reduced infrastructure MTTR by automatically resolving high CPU/Memory events and clearing false-positive alerts using anomaly telemetry.

Infrastructure EKS Provisioner

Terraform · AWS EKS · VPC · IAM · IRSA | [GitHub Link](#)

- Designed modular Infrastructure-as-Code (IaC) to provision a production-grade, multi-AZ AWS environment with private subnets and NAT gateways.
- Deployed a zero-trust architecture by enforcing least-privilege network isolation and eliminating long-lived credentials via IAM Roles for Service Accounts.

High-Throughput FastAPI Microservice

Python · FastAPI · MongoDB · Docker | [GitHub Link](#)

- Developed a production-grade, asynchronous REST API optimized for non-blocking I/O and high-concurrency data ingestion.
- Implemented strict schema validation using Pydantic and containerized the backend architecture to ensure rapid, domain-agnostic deployment.

EDUCATION

Master of Science in Computer Science (Specialization in Cybersecurity)

Troy University, Alabama, USA | Jan 2024 – Jun 2025

- Thesis: Smart Contract Vulnerability Scoring Framework. Extracted foundational security KPIs to develop a mathematical model that evaluates and quantifies code vulnerabilities.
- Achievement: Winner (1st Place), TroyHack 2024. Collaborated within a 4-person team to rapid-prototype an educational 3D simulation game (Unity/C#).

Advanced Certification in Data Science

IIIT Bangalore, India | Mar 2023 – Nov 2023

- Focus: Machine Learning, Predictive Analysis and Python (GPA: 3.7/4.0)

Bachelor of Science (Electronics)

Guru Ghasidas Central University, India | Jun 2018 – Sep 2021

- Capstone: Greenhouse Control & Management System using PIC Microcontrollers and sensor network.